PC-AVCT: Context-Augmented, Session-Calibrated Blood Pressure Tracking from PPG

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Abstract—Cuffless blood pressure (BP) estimation from photoplethysmography (PPG) is vulnerable to participant leakage, ambiguous calibration burden, and protocol shortcuts. We evaluated PC-AVCT, a context-augmented implementation motivated by Attractor–Vascular Coupling Theory (AVCT), for session-anchored systolic (SBP) and diastolic (DBP) tracking. The primary retrospective analysis used three repeats of five-fold participant-level internal cross-validation on 653 AuroraBP participants, excluding calibration-labelled rows from fitting and scoring. With a three-reading cuff anchor, the rerun wearable-state model achieved participant-level MAE of 7.697/6.139 mmHg. Paired gains over the same core without added context were 0.436 mmHg SBP and 0.126 mmHg DBP (Holm-adjusted paired-test $p < 0.001$ for both). In a separate 108-participant pulse-rate study, quality-weighted visible/infrared fusion achieved 3.690 BPM MAE. A post-hoc comparison against equal weighting (3.985 BPM) showed a 0.294 BPM improvement (0.069–0.567); guarded skin-tone correction did not improve overall error. A separate VitalDB implementation, fitted using 55 development subjects, was evaluated on 228 cases from 227 disjoint held-out subjects and 5,472 future windows. A fixed 50/50 blend of context and context-plus-PPG predictions improved over calibration, time, and demographics by 0.451 mmHg SBP (0.121–0.781) and 0.357 mmHg DBP (0.194–0.512), reaching 15.375/8.833 mmHg. These results support incremental context and PPG contributions in the evaluated settings, with separate evidence for pulse-rate fusion. They do not isolate quality-dependent BP weighting, demonstrate transfer of the fitted AuroraBP model, or establish cuff replacement, equitable performance, prospective validation, or standards compliance.

Index Terms—Photoplethysmography, cuffless blood pressure, nonlinear time-series features, one-session calibration, quality-aware fusion, pigmentation-intervention audit, data leakage, participant-level cross-validation, wearable sensing.

I. Introduction

A. Motivation

PPG-based cuffless BP estimation is attractive because optical sensors are already present in consumer wearables. Its evaluation must distinguish participant overlap, calibration burden, and acquisition-context shortcuts. Record-level splits can inflate apparent accuracy when correlated measurements from one participant enter both training and testing [1]. A model anchored to three initial cuff readings also answers a different question from one calibrated once or repeatedly supplied with recent cuff values. Finally, pigmentation-associated optical differences motivate intervention testing, but a small subgroup or a nonsignificant contrast cannot establish equitable performance [2]–[4]. We audit anchoring, context, and pigmentation-related interventions using recorded Fitzpatrick labels. Candidate interventions and context arms were frozen before repeated participant-level validation, after earlier development on AuroraBP. Because a fixed holdout was inspected across successive ablations, it is retained as historical development evidence. The primary repeated analysis confines feature ranking and fitting to outer training participants but remains internal validation. Secondary negative results are reported without interpreting them as proof of equity, equivalence, or general failure of pigmentation-aware modeling.

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B. What "PC-AVCT" Names

We retain PC-AVCT for continuity with the author-developed AVCT formulation [5]. Here it denotes a context-augmented, session-calibrated tracking framework with a secondary pigmentation-intervention audit. Three implementations must be distinguished: AuroraBP evaluates added context, the multi-wavelength study evaluates pulse-rate fusion, and VitalDB evaluates a separately fitted context-plus-PPG model and a fixed blend. Quality-dependent wavelength weighting is implemented in the pulse-rate study; it is not the VitalDB blending rule. The optional MST-dependent pulse-fusion route uses an exploratory inner-validation guard. These implementations do not constitute a single unchanged model transported between datasets.

C. Hypotheses

The development program considered five hypotheses, summarized here with a narrowed anchor-specific H5. This revised wording is retrospective and must not be interpreted as a prospectively specified hypothesis set. Historical model-development ablations used a fixed 501/168 split; the frozen final comparison was subsequently assessed by repeated participant-level outer cross-validation:

- H1 A one-session calibration anchor, combined with AVCT attractor features, tracks within-session SBP/DBP change more accurately than calibration retention (predicting no change).
- H2 Acquisition-context covariates improve tracking under a session anchor, with a separately tested conservative arm restricted to wearable-available posture, activity, and optical quality.

- H3 Pigmentation-corrected optical features and explicit Fitzpatrick-aware model terms improve accuracy and/or narrow the between-skin-tone error gap relative to a reference model.
- H4 Rebalancing the training population across Fitzpatrick bands improves fairness (narrows the between-band MAE gap) without harming overall accuracy.
- H5 A same-session "most-recent-prior" cuff anchor can account for a large part of apparent improvement relative to session-start anchoring.

H1 and H2 are supported for the evaluated models, and the historical anchor comparison supports the narrower H5. These comparisons do not isolate the attractor feature bank from all other inputs. The specific interventions tested under H3 and H4 are not supported: overall pigmentation-correction benefit is not established and the combined rebalancing/group-gating intervention increases error in the historical matched ablation. The small Fitzpatrick V–VI subgroup prevents interpreting those null or adverse intervention results as a general falsification of pigmentation-aware modeling or as evidence of equity.

D. Contributions

1) Participant-separated internal evaluation of added wearable-state context under a three-reading session anchor, with a protocol-enriched comparator and matched calibration-dose analysis.
2) A calibrated-tracking audit that separates initial calibration from recent-cuff anchoring and reports secondary pigmentation-intervention results without fairness claims.
3) Independent pulse-rate fusion evaluation and a separately fitted VitalDB implementation that tests incremental PPG utility beyond calibration, time, and demographics. The equal-weight pulse comparison was added post hoc during manuscript audit.

The main text presents the primary comparisons and independent-dataset results. Historical development ladders, auxiliary diagnostics, and exploratory endpoint specialization are retained in the supplementary appendices.

II. Related Work

A. PPG-Based Cuffless Blood Pressure Estimation

Cuffless BP estimation from PPG spans pulse-transit-time methods, hand-engineered waveform features, and end-to-end deep networks [6], [7]. Calibration-free approaches attempt to avoid any subject-specific reference [8], typically at a cost in accuracy relative to calibrated methods; single-point-calibrated approaches instead accept one reference reading and model change from it [5], [9]. Recommendations for evaluating PPG-based BP algorithms increasingly emphasize AAMI/ISO 81060-2 and BHS grading protocols and honest reporting of what these standards do and do not certify for calibrated, per-participant models [10].

B. Nonlinear and Information-Theoretic Feature Representations

The feature bank rests first on established time-series methods. Delay-coordinate reconstruction provides a generic mathematical basis for representing state from a scalar observation under specific dynamical assumptions [11]; it does not by itself prove that a finite PPG window recovers a unique cardiovascular state. Recurrence plots and recurrence-quantification analysis provide established descriptors of reconstructed trajectories [12], [13], while sample entropy and permutation entropy were developed as general complexity measures, including for short physiological series [14], [15]. PPG itself reflects cardiac-synchronous peripheral blood-volume change modulated by vascular, autonomic, respiratory, thermoregulatory, contact, and optical factors [16]. Independent work has specifically applied attractor reconstruction to arterial pulse-wave morphology [17].

CST, AVCT, and ADT are recent author-developed, hypothesis-generating formulations layered on these established methods [5], [18], [19]. They motivate the particular feature grouping, CSI term, calibration-delta representation, and terminology used here, but they are not treated as independently validated physiological laws, and their axioms are not required for the statistical validity of the empirical comparisons. Accordingly, this study tests whether the resulting feature architecture generalizes under participant-level validation; it does not claim that successful prediction proves the theories. A residualized mutual-information ranking restricts the candidate feature set rather than searching combinatorially over all features [20], [21].

C. Skin Pigmentation and PPG-Derived Vitals

Pigmentation-associated optical differences motivate evaluating both signal processing and downstream error [2]–[4], [22], [23]. A Nigerian rPPG field study enrolled 306 Fitzpatrick V/VI participants, obtained 249 analyzable estimates, and reported 15.4/10.9 mmHg SBP/DBP MAE with poor hypertension sensitivity [24]. Its lack of an I–IV comparator prevents attributing error specifically to skin tone. A small polarization-sensitive PPG study improved perfusion index in eight analyzed participants but did not estimate BP [23]. Such optical findings and gain-calibration methods [25] motivate interventions; they do not establish BP-tracking benefit. We therefore distinguish signal-quality associations, pulse-rate fusion, and BP error, and avoid interpreting nonsignificant subgroup differences as parity.

D. Evaluation Rigor and Data Leakage

The original AuroraBP study already emphasized strong calibration-based benchmarks and differences between laboratory and ambulatory performance [26]. The present contribution is the specific context, calibration-dose, and intervention comparisons on the analyzed subset; it is not the introduction of calibration-aware benchmarking itself.

A benchmark study across four public PPG-BP datasets (Sensors, UCI, BCG, PPGBP) explicitly compares record-level-leaked and subject-independent splits and reports that leakage alone can inflate apparent performance by more than 40 percentage points on some metrics [1], corroborating a broader concern in the deep-learning BP literature that record-level or window-level splits can substantially inflate reported accuracy relative to subject-independent evaluation. The MIMIC-BP data descriptor addresses this reproducibility problem by providing 1,524 subject-identified cases, balanced per-participant segments, and participant-disjoint training, validation, and test lists [27]; it is cited as a relevant external resource but was not used in the primary or endpoint-specialization results reported here. A separate line of work interrogates the information content of the PPG waveform itself, asking how much of reported model accuracy reflects genuine physiological signal versus dataset-specific shortcuts [28], [29]. Our calibration-row leakage audit (Section III-E) is a distinct but related artifact: not a train/test split violation, but a same-session near-zero-delta artifact specific to change-prediction (rather than absolute-value) BP models, and we are not aware of a prior study that isolates it.

III. The PC-AVCT Framework

A. Attractor Feature Bank

Let x_n, $n = 1, \ldots, T$, denote a discretely sampled PPG window at rate f_s. Following delay-coordinate embedding [11], we reconstruct the trajectory

$$\mathbf{X} = \tau_{\mathcal{G}}[x], \qquad \mathbf{X}_{n,\cdot} = [x_n, x_{n+\tau}, \ldots, x_{n+(d-1)\tau}], \tag{1}$$

with embedding dimension $d = 4$ and delay $\tau = 5$ samples, matching the AVCT/CST instantiation [5], [18]. From $\mathbf{X}$ we compute the shared PPG attractor feature bank

$$\begin{aligned}\Phi(x) = [\,&\sigma_m,\ \mathrm{skew}_m,\ \mathrm{kurt}_m,\ H_{\mathrm{SE}},\ H_{\mathrm{PE}},\ R_{\mathrm{rec}},\\ &R_{\mathrm{det}},\ H_{\mathrm{RQA}},\ \mathrm{CSI},\ H_{\mathrm{spec}},\ Q\,]\end{aligned} \tag{2}$$

comprising waveform moments (σ_m, skewness, kurtosis), sample and permutation entropy (H_{SE}, H_{PE}), recurrence rate and determinism (R_{rec}, R_{det}), recurrence entropy (H_{RQA}), CSI, spectral entropy, and optical quality. The auxiliary stress-test bank labels the unmodified measured waveform ref_*, the simulated stressed waveform raw_*, and its corrected version pc_*; "reference" does not mean latent physiological ground truth. AuroraBP instead extracts raw_* from observed optical waveforms and pc_* after the correction below. It does not construct an AuroraBP ref_* input. The primary candidate pool contains raw-feature changes, time variables, and a UCI-derived prior; its exact provenance is described in Section IV-B.

B. Pigmentation Correction Operator

The implemented AuroraBP operator is a heuristic motivated by attenuation, rather than an estimate of melanin or a fitted Beer–Lambert law. Let $z = (\mathrm{FST} - 1)/5$, $b = \mathrm{median}(x)$, $a = \max(0.45, 1 - 0.55z)$, and $u = (x - b)/a$. With $W(u)$ a Wiener-filtered version of u and $\lambda = \mathrm{clip}(0.75z, 0, 0.75)$, the corrected channel is $b + (1 - \lambda)u + \lambda W(u)$. The filter kernel is an odd number of samples derived from $0.05 f_s$, bounded between 5 and 21. A residual-to-bandpassed-signal SD ratio, clipped to $[0, 1.5]$, modifies entropy/recurrence tolerances; the corrected channel uses 0.35 times that ratio. The waveform median is a baseline proxy, not a separate measured DC channel. In the repeated participant-CV ablation, corrected-minus-raw MAE was -0.007 mmHg SBP (95% CI -0.042 to 0.029) and -0.030 mmHg DBP (-0.062 to 0.003). These intervals do not establish overall benefit or equivalence. The ablation tests the complete correction-and-feature-extraction configuration, not attenuation rescaling alone.

C. Calibration Dose and Anchoring

For participant p, let $b_p^{(k)}$ denote an anchor formed from exactly k initial cuff readings ($k \in \{1, 2, 3\}$), with $k > 1$ using their arithmetic mean. We define the session-anchored target

$$\Delta_{\mathrm{session}}^{(k)}(p, i) = y_{p,i} - b_p^{(k)}, \tag{3}$$

where $y_{p,i}$ is the i-th subsequent cuff reading for participant p, and the prior-anchored target

$$\Delta_{\mathrm{prior}}(p, i) = y_{p,i} - y^{*}_{p,i-1}, \tag{4}$$

where $y^{*}_{p,i-1}$ is the most recent valid prior measurement (which may itself be a tracked, non-calibration reading). These are not the same question. Eqn. 3 evaluates BP change after an initial calibration session without supplying later cuff values to the predictor. Eqn. 4 asks whether PPG can track change from the most recent cuff reading, which in this in-lab protocol can be only minutes old. A model evaluated under Eqn. 4 solves an easier problem; 83% of the historical fixed-split gain under prior anchoring was attributable to anchor replacement alone. We therefore lead with session-anchored tracking results and explicitly report calibration dose; neither anchoring scheme establishes cuff replacement. "One reading" refers only to $k = 1$; the primary $k = 3$ analysis is described as one-session calibration using three readings. The calibration-retention baseline predicts $\hat{\Delta} = 0$ under either anchor, i.e., "the BP has not changed since calibration (or since the last reading)." This is one baseline; the main context contrasts compare fitted models, and VitalDB additionally uses a calibration-plus-context control.

D. Context Operator

Let $c(p, i)$ collect acquisition context. We distinguish a wearable-state set (posture, activity, and optical signal quality) from a protocol-enriched set that additionally includes protocol phase, recording duration, and pressure-quality metadata. The wearable-state variables require corresponding sensing or classification at deployment; the protocol-enriched variables may encode AuroraBP acquisition structure and are therefore interpreted as an internal upper bound on context utility. The context-augmented model is

$$\hat{\Delta}(p,i) = f_\theta\big(\Phi(x_{p,i}),\, \Phi(x_{p,i}) - \Phi(x_{p,\text{anchor}}),\, c(p,i)\big), \quad (5)$$

where f_θ is a gradient-boosted tree ensemble (HistGradientBoosting; hyperparameters grid-searched over learning rate, maximum leaf nodes, and ℓ_2 regularization) trained with either squared or absolute loss (Section IV), and the first-difference term $\Phi(x_{p,i}) - \Phi(x_{p,\text{anchor}})$ is the attractor-feature analogue of Eqns. 3–4 in feature space.

E. Calibration-Row Leakage Audit

A specific artifact threatens any context-augmented change-prediction model: a calibration-measurement indicator in $c(p,i)$ could identify trivial rows if the target reading were compared with itself; a calibration label alone does not guarantee this condition. A model that learns "calibration row ⇒ predict no change" gains accuracy on those rows for free, in a way that would not transfer to genuine tracking of a new, non-calibration measurement. We formalize four checks:

A Compare $|\Delta|$ on calibration-flagged vs. unflagged rows – are flagged rows trivially near-zero?
B Re-score the same fitted models with flagged rows removed from the holdout only.
C Refit with the calibration-flag feature removed, flagged rows still scored.
D (Decisive) Refit with the flag feature removed and flagged rows excluded from both fitting and scoring – fit and evaluation restricted consistently.

Check D is decisive because A–C each leave one channel open (A is diagnostic only; B changes the scored population without changing the fitted model; C changes the model without changing the scored population). If the context gain measured under Check D remains close to the unaudited figure, the effect is not an artifact of the calibration flag. Section B-F reports all four.

Finding 1. On this dataset, calibration-flagged rows are not near-zero-delta (mean $|\Delta|$ = 6.774 vs. 9.902 mmHg SBP on development rows − smaller, but not trivially so), and Check D confirms the context gain survives essentially intact (+0.614 vs. the unaudited +0.809 mmHg SBP). Leave-one-out results implicate recording duration and post-exercise activity rather than the calibration flag (Section B-F).

IV. Methodology

A. Datasets

AuroraBP (primary dataset) is the released Aurora-BP study resource [26], [30]. The files used here provide 1,221 participant records and 11,597 auscultatory measurements across in-lab calibration, static, exercise, and temporal-challenge protocol phases, with real Fitzpatrick skin-type labels (I: 311, II: 294, III: 138, IV: 44, V: 29, VI: 8 participants; 397 unlabeled). Waveforms were resolvable for 672 participants. The calibration-dose comparison uses the common subset of 653 participants with three valid initial calibration readings and 9,216 matched post-initial-calibration rows; the strict deployment endpoint excludes 1,337 later calibration-labelled rows, leaving 7,879 scored measurements. Historical ablations used a fixed 501/168 participant split (Fitzpatrick composition of the fixed holdout: I–II 133, III–IV 29, V–VI 6). Because that holdout was inspected repeatedly during model development, it is retained as historical development evidence rather than described as untouched validation. Four secondary sources appear in the development notebooks: BIDMC (53 subjects; 1,272 windows) [31]–[33], sampled MIMIC-II (1,500 windows) [34], [35], Welltory-PPG (106 windows) [36], and UCI (425 records; 3,268 windows) [37], [38]. BIDMC, MIMIC-II, and Welltory form a historical corrected-feature PCA bank; the primary candidate list does not include those PCA coordinates. UCI trains the BP-change prior that can enter the primary model. Synthetic pigmentation ranks on these auxiliary sources are simulation inputs, not measured participant traits, and support no equity claim.

TABLE I
AuroraBP Cohort and Strict-Endpoint Flow

Stage	Participants/measurements
Released participant records	1,221
Released auscultatory measurements	11,597
Participants with resolvable waveforms	672
Common three-reading calibration cohort	653
Matched post-initial-calibration rows	9,216
Later calibration-labelled rows excluded	1,337
Strict-endpoint scored rows	7,879

Two additional datasets are used for independent follow-up analyses. The Appel–Theart Participant Study Data provide paired visible and infrared PPG, Monk Skin Tone (MST), and repeated pulse-rate references for 110 participant folders [39]. After waveform and metadata auditing, 108 participants contributed 630 analyzable pulse-rate windows, including 24 participants with MST 7–10. Each participant has only one cuff BP value; consequently BP analysis on this dataset is explicitly cross-sectional and does not test calibration tracking. VitalDB contains synchronized intraoperative waveforms and clinical tracks from 6,388 cases [40], [41]. The first 60 deterministic eligible-case positions yielded 55 development subjects. The next 240 positions yielded 228 eligible cases from 227 disjoint held-out subjects and 5,472 future windows after waveform/reference quality control. One held-out subject contributed two cases; each eligible case contributed 24 scored windows. Subject identities do not overlap development and holdout, and all cases belonging to a subject are grouped for inference. Only SNUADC/PLETH, Solar8000/ART_SBP, and Solar8000/ART_DBP were requested; ECG was neither loaded nor used. VitalDB has no skin-tone field. It evaluates future-window tracking within a separately developed intraoperative implementation, not pigmentation fairness or transfer of AuroraBP-fitted parameters.

B. Feature Extraction and Selection

AuroraBP features are extracted from the central 10-s optical window at $f_s = 500$ Hz (5,000 samples), using a third-order 0.5–8 Hz bandpass, $d = 4$, and $\tau = 5$ samples (10 ms). The extractor calculates $f_s = (N-1)/(t_{N-1} - t_0)$ from the recording timestamps; all 11,427 cached feature rows match the timestamp audit at 500 Hz within 2×10^{-11} Hz. The central waveform is not resampled before filtering. Sample entropy uses order 2, a base standardized tolerance of 0.2, and at most 600 samples; permutation entropy uses order 3. Recurrence analysis uses at most 300 standardized embedded points, Chebyshev distance, a base radius of 0.2 modified by the noise ratio, and diagonal lines of length at least 2. For the filtered AuroraBP signal, let $S(f)$ be its Welch power spectral density, with 4-s Hann segments and 50% overlap. The quality feature is

$$Q = \frac{\sum_{0.5 \le f \le 5} S(f)}{\sum_{0.1 \le f \le 15} S(f)}. \tag{6}$$

CSI is $0.4R_{\text{det}} + 0.3[1 - \text{clip}(H_{\text{RQA}}/5, 0, 1)] + 0.3Q$. Recalculation from the cached components agrees with stored CSI to 2.3×10^{-16}. This derived spectral ratio, stored as raw_ppg_quality, is distinct from AuroraBP's supplied optical_quality context field and the independent pulse-fusion weights q_V, q_I. The historical hybrid calibration-anchored feature augmentation is denoted HCA below; its source-code identifiers retain the older name H1, distinct from hypothesis H1.

Within each outer training fold, the primary code ranks raw-feature differences and relative differences, elapsed-time variables, and the UCI prior after ridge residualization against baseline BP and log elapsed time. It applies mutual_info_regression to the residuals and retains up to 24 candidates plus baseline BP. SMD and AUC were historical diagnostics, not the primary selection algorithm. The name "no context" means no added context variables: baseline BP, candidate elapsed-time terms, and the UCI prior can still enter its shared core. No ref_* feature enters that core. The UCI prior is generated from corrected features, so absence of explicit Fitzpatrick terms does not prove absence of indirect Fitzpatrick dependence. The regenerated selection manifests retain the UCI prior in 26 of 30 endpoint/fold core models; all context arms share the corresponding selected core. Historical SD ratios are reported in Table S4; they are not variance fractions.

C. Model and Training

For the primary AuroraBP comparisons, the supplied implementation uses median imputation followed by HistGradientBoosting with squared-error loss, learning rate 0.03, 15 maximum leaf nodes, ℓ_2 regularization 1.0, 300 iterations, and random seed 42. These settings are inherited from development rather than reselected using outer test folds. Row weights are inverse participant measurement counts, clipped at the training-set 2nd/98th percentiles and normalized to mean one. Participant folds are stratified by Fitzpatrick band, with seeds $42 + 26200 + r$ for zero-based repeat r. The change gate is selected using fitted training predictions, not an independent inner validation set; reported activation is 1.0. This training-stage gate selection is an additional development limitation. Exploratory endpoint specialization uses a separate nested procedure.

D. Evaluation Protocol

Primary results use three repeats of five-fold participant-stratified outer cross-validation over the common 653-participant cohort. Every participant appears in an outer test fold once per repeat. Feature ranking, imputation, model fitting, and gate selection are repeated using only the corresponding outer training participants; no record or window from an outer test participant enters training. The final feature specification and context arms were frozen before this repeated analysis. Calibration-labelled rows are excluded from both fitting and scoring. Per-participant MAE is averaged across repeats, and 95% CIs are obtained by participant bootstrap (10,000 resamples); context contrasts use paired per-participant differences. This is repeated internal validation because AuroraBP informed earlier model development, not an external or prospective test. Historical fixed-split analyses use participant bootstrap (4,000 resamples) and are labelled accordingly. BHS and AAMI/ISO results are retained as indicative historical context, not compliance claims.

1) Claim hierarchy and multiplicity: The two primary retrospective comparisons are SBP and DBP wearable-state context versus the same core without added context. The rerun saves each participant's mean MAE across the three repeats and forms paired no-context-minus-wearable-state contrasts. Two-sided paired sign-flip tests use the absolute mean contrast, 99,999 random sign assignments (seed 20260910), and $(b+1)/(B+1)$ Monte Carlo p values. Holm's step-down procedure controls multiplicity across these two tests at $\alpha = 0.05$ [42]. Paired t tests are reported as a sensitivity analysis. No p value is reconstructed from a CI width. Primary contrasts additionally report 97.5% participant-bootstrap intervals, giving approximate Bonferroni simultaneous 95% coverage for the two endpoints; other intervals are marginal 95%. The sign-flip test assumes sign exchangeability under the paired null. Inference conditions on the fitted out-of-fold predictions, treats participants as the sampling units, and does not capture all dependence from shared training sets or earlier model development. Multiplicity correction does not make this retrospectively designated family prospective confirmation. All other comparisons are secondary or exploratory. For the fixed-estimator pigmentation diagnostic, the prior-anchored reference model was refitted on the original 501 development participants and evaluated on the same 168 held-out participants. Fitzpatrick bands stratify the resulting predictions; they do not alter the fitted estimator or its gates. The upstream corrected-feature UCI prior can introduce indirect Fitzpatrick dependence. MAE is averaged within participant

before band means, 10,000-resample bootstrap intervals, and exploratory between-band contrasts are computed. A separate band summary uses the regenerated primary wearable-state predictions from all 653 participants. These diagnostics cannot establish fairness, particularly with only 6 and 25 V–VI participants in the respective cohorts. The later endpoint-specialization analyses are explicitly exploratory. They use the same 653-participant strict cohort and three repeats of five-fold participant outer validation, with three-fold participant inner validation for candidate selection. For SBP, the candidates are ridge- or spline-based signed-residual corrections, shrunk and clipped using inner-fold choices; for DBP, the default candidate is the unchanged base predictor. The exploratory DBP router predicts whether $|\Delta_{\mathrm{DBP}}|$ exceeds a training-fold-derived 80th-percentile cutoff. Its threshold is selected without outer-fold labels, and its output does not alter DBP: it only requests repeat measurement or recalibration. We report discrimination, calibration, warning precision and recall, and warning rate. Because these analyses followed inspection of earlier AuroraBP results, their confidence intervals quantify internal resampling stability rather than independent confirmation.

For the independent multi-wavelength mechanism analysis, visible and infrared pulse-rate estimates are weighted by within-window spectral concentration and autocorrelation quality. The tone-blind optical arm uses no MST information. The guarded arm permits an MST-dependent wavelength-weight exponent selected within four participant-disjoint inner folds. It activates only when the proposed nonzero exponent improves mean inner-fold MAE by at least 0.10 BPM and improves all inner folds; otherwise it returns exactly to tone-blind optical fusion. Evaluation uses 20 repeats of five-fold participant outer cross-validation. The supplied script averages repeated out-of-fold predictions for each window before calculating participant MAE; these are repeated-cross-fit ensemble estimates, not mean per-repeat errors. The audit's fixed equal-weight comparator is $(h_V+h_I)/2$, requires no fitting, and uses exactly the same 630 windows. Paired differences are averaged within each of 108 participants and bootstrapped 10,000 times (seed 20260910); those additional comparisons are post hoc with marginal intervals.

For VitalDB, the median BP and median features of three early valid 20-s windows form the case-specific calibration anchor and are excluded from fitting and scoring; up to 24 later windows per case are scored. A calibration-plus-context control uses the anchor value, elapsed time, age, sex, and preoperative hypertension. The PPG arm adds changes in pulse rate, spectral concentration and entropy, waveform and derivative energy, distributional shape, beat-interval variability, and signal quality. After the initial 55-subject pilot, this stricter PPG-versus-calibration-plus-context comparison was frozen for the expanded test; it was not prospectively registered. Candidate estimators, regularization, clipping, and the context/PPG blend were selected using only the 55 development subjects. The selected models were then fitted on development data and applied once to 228 cases from 227 disjoint held-out subjects. Both endpoints selected a constant PPG weight of 0.5: the final prediction is the arithmetic mean of context and context-plus-PPG predictions. Signal quality is an input, not a window-dependent blend weight. This is a separately fitted implementation, not an unchanged AuroraBP model transferred to VitalDB. Reference quality control required plausible SBP/DBP, local arterial stability, and pulse pressure; it did not use prediction error. Contrasts were averaged within subject before a 10,000-resample participant bootstrap. The holdout was not used for scaling, model selection, clipping, or blend selection.

TABLE II
Refitted Repeated Participant-Level Validation (653 Participants; Three-Reading Anchor)

Arm	SBP MAE [95% CI]	DBP MAE [95% CI]
Calibration retention	9.790	6.767
No added context	8.133 [7.824, 8.455]	6.265 [6.019, 6.526]
Optical quality only	8.110 [7.802, 8.431]	6.273 [6.025, 6.535]
Wearable-state context	7.697 [7.407, 8.008]	6.139 [5.899, 6.391]
Protocol-enriched upper bound	7.523 [7.243, 7.815]	5.743 [5.527, 5.977]
Primary gain [97.5% CI]	0.436 [0.305, 0.573]	0.126 [0.053, 0.200]
Holm-adjusted paired p	0.00002	0.00013

V. Experiments and Results

A. Primary Repeated Participant-Level Validation

Table II reports the refitted comparison on 653 participants and 7,879 strict-endpoint measurements. The wearable-state arm achieves 7.697/6.139 mmHg SBP/DBP. Its paired gains over no added context are 0.436 and 0.126 mmHg. The corresponding 97.5% bootstrap intervals are $[0.305, 0.573]$ and $[0.053, 0.200]$ mmHg. Holm-adjusted paired sign-flip p values are 0.00002 and 0.00013; paired t sensitivity tests also pass the two-endpoint Holm threshold. These conditional internal-validation results support the added context comparison. Posture and activity are protocol annotations; deployment requires independent wearable estimates. The protocol-enriched arm additionally uses phase, duration, and pressure-quality metadata and is interpreted as an internal upper bound.

B. Calibration Dose

Table III reports freshly refitted protocol-enriched models on identical participants and future measurements for each calibration dose. Ranking and fitting are repeated within the same participant folds. These dose gains belong to the protocol-enriched arm, not the lead wearable-state arm. Three readings are one-session repeated calibration, not one-reading calibration.

C. Prior-Anchored Isolation Ladder: Where the 26.2% Figure Really Comes From

Table IV decomposes the strongest raw number in this study – a 2.084 mmHg (26.2%) SBP reduction – into its

TABLE III
Refitted Protocol-Enriched Models by Calibration Dose (653 Participants)

Target	Readings	Retention	Model MAE [95% CI]	Gain vs. retention
SBP	1	10.547	8.257 [7.933, 8.602]	2.290 [2.051, 2.525]
SBP	2	10.037	7.750 [7.458, 8.055]	2.286 [2.066, 2.509]
SBP	3	9.790	7.523 [7.243, 7.815]	2.267 [2.044, 2.489]
DBP	1	7.213	6.280 [6.026, 6.548]	0.934 [0.739, 1.129]
DBP	2	6.900	5.873 [5.648, 6.112]	1.027 [0.833, 1.215]
DBP	3	6.767	5.743 [5.527, 5.977]	1.024 [0.832, 1.211]

TABLE IV
Historical Controlled Isolation Ladder, Prior-Anchored (SBP, common dev-only feature set, 168 holdout participants)

Step	MAE	Gain [95% CI]	Share
Session anchor (legacy)	7.956	–	–
→ Prior anchor	6.216	+1.739 [1.296, 2.197]	83%
→ Absolute-error loss	6.114	+0.102 [−0.028, 0.240]	5%†
→ +Context	5.795	+0.319 [0.189, 0.446]	15%
→ +HCA features	5.872	−0.077 [−0.183, 0.024]	−4%†
Total (session → full)		+2.084 [1.637, 2.533]	26.2%

† CI crosses zero: not established at 95%.

components under a controlled, feature-set-fixed ladder. Eighty-three percent of that gain is attributable to re-anchoring alone (predicting from the most recent prior reading rather than the session calibration), which on an in-lab protocol can mean a reference only minutes old; this is a genuine tracking result, but it is not a cuffless-replacement result, and the two must not be reported interchangeably (Section III-C).

D. Secondary Pigmentation-Intervention Audit

Table V compares four configurations, all built on the prior-anchored absolute-error+context base model of Table IV (chosen because it is the strongest available base for isolating the marginal effect of skin-tone modeling): reference (no added Fitzpatrick terms; indirect dependence is possible), skin-features-only (continuous Fitzpatrick terms and interactions, no rebalancing), skin-features+balance (adds participant rebalancing across Fitzpatrick bands and group-specific shrinkage gates learned on development data), and updated-skin-aware (the full configuration). All four share participants, targets, holdout, base features, estimator, and hyperparameters. Skin features alone show no established benefit (5.770 vs. 5.795 mmHg SBP reference, not CI-distinguishable). Adding the combined rebalancing and group-gating intervention moves SBP MAE from 5.770 to 6.086 mmHg, a point difference of 0.316 mmHg. A paired interval for this specific incremental contrast is not supplied; the reported 0.291 mmHg interval instead compares the full intervention with the original reference arm. These experiments do not isolate rebalancing from group gating. In the signed contrast reported in Table V, reference minus skin-aware is therefore −0.291 mmHg $[-0.420, -0.170]$. The combined intervention also widens the between-Fitzpatrick-band MAE gap, from 0.556 to 0.772 mmHg (SBP) and 0.721 to 0.940 mmHg (DBP). Per-band results suggest harm in the two better-represented

TABLE V
Historical Fixed-Split Skin-Tone Model Comparison (168 Evaluation Participants)

Configuration	SBP MAE	DBP MAE	vs. blind
Reference	5.795	4.486	–
Skin-features-only	5.770	4.477	Neutral
+ Demographic balance	6.086	4.712	Harm
Updated skin-aware (full)	6.086	4.712	Harm
Paired diff. (reference − full), mmHg [CI]; negative favors reference:			
SBP	−0.291 [−0.420, −0.170]		
DBP	−0.226 [−0.345, −0.116]		
Between-band gap, reference (SBP/DBP)	0.556 / 0.721 mmHg		
Between-band gap, skin-aware (SBP/DBP)	0.772 / 0.940 mmHg		

TABLE VI
Pigmentation-Corrected vs. Raw Channel in Repeated Participant CV

Target	Scope	n	Raw	Corrected	Δ MAE [95% CI]
SBP	Overall	653	7.524	7.517	−0.007 [−0.042, 0.029]
SBP	I–II	516	7.487	7.482	−0.005 [−0.041, 0.032]
SBP	III–IV	112	7.683	7.620	−0.064 [−0.168, 0.036]
SBP	V–VI	25	7.571	7.793	0.222 [0.043, 0.401]
DBP	Overall	653	5.743	5.713	−0.030 [−0.062, 0.003]
DBP	I–II	516	5.681	5.639	−0.043 [−0.079, −0.005]
DBP	III–IV	112	6.064	6.077	0.014 [−0.066, 0.104]
DBP	V–VI	25	5.578	5.602	0.024 [−0.084, 0.131]

Δ = corrected minus raw; negative values favor correction.

bands; the Fitzpatrick V–VI comparison is indeterminate and rests on only six evaluation participants. These historical fixed-split results do not support adopting the tested combined intervention but do not establish general fairness properties.

The supplementary subgroup table now reports newly refitted historical-method estimates alongside the regenerated primary wearable-state results. The companion change record documents the differences from the superseded subgroup values.

1) Repeated-CV pigmentation-correction intervention: Table VI reports the separate strict-cohort repeated-CV ablation. Neither overall interval establishes a benefit; equivalence was not tested. The V–VI SBP subgroup shows a CI-supported 0.222 mmHg worsening under correction, whereas the I–II DBP subgroup shows a small CI-supported improvement. These subgroup findings are heterogeneous and exploratory; they do not support a general skin-aware-model benefit. Adding explicit Fitzpatrick interactions to the corrected channel did not establish overall benefit: SBP $\Delta = 0.001$ mmHg (95% CI −0.042 to 0.041) and DBP $\Delta = -0.037$ mmHg (−0.076 to 0.002), relative to the raw model.

An exploratory signal diagnostic collapsed all repeated observations to one median per participant before inference. Fitzpatrick type was associated with lower raw-channel PPG quality ($\rho = -0.157$, 95% CI −0.234 to −0.079), greater normalized attractor dispersion ($\rho = 0.511$, 0.447 to 0.571), and lower recorded optical-quality score ($\rho = -0.174$, −0.253 to −0.092). Associations with optical noise ratio ($\rho = 0.032$, −0.044 to 0.112)

TABLE VII
Independent Multi-Wavelength PPG Mechanism Test

Arm	Overall MAE (BPM)	MST 7–10 MAE
Visible only	4.366	5.434
Infrared only	5.037	5.165
Equal-weight fusion (post hoc)	3.985	4.587
Quality-weighted fusion	3.690	4.184
Guarded MST-aware fusion	3.703	4.265

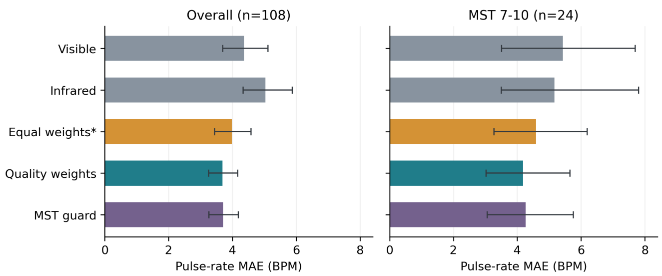

Fig. 1. Participant-balanced pulse-rate MAE with marginal 95% bootstrap intervals. The equal-weight comparison was added post hoc; quality weighting improves overall error relative to equal weighting, while guarded MST correction does not establish additional benefit.

TABLE VIII
VitalDB Context-Plus-PPG Frozen Holdout (227 Subjects)

Target	Retention	Context	Context+PPG	PPG−Context [95% CI]
SBP	25.291	15.827	15.375	−0.451 [−0.781, −0.121]
DBP	13.098	9.189	8.833	−0.357 [−0.512, −0.194]

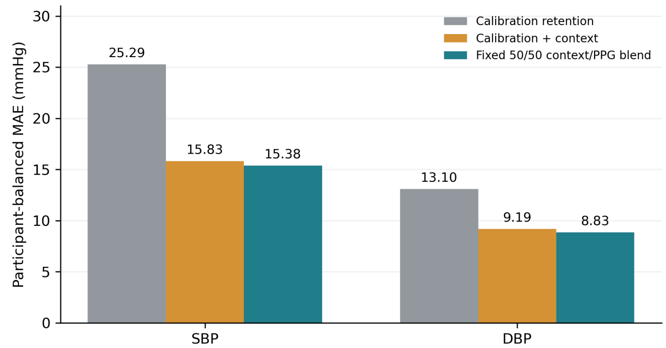

Fig. 2. VitalDB locked-holdout participant-balanced MAE. Calibration-plus-context explains most of the gain over retention; the frozen fixed context/PPG blend adds small CI-supported incremental improvements for both endpoints.

and raw spectral entropy ($\rho = 0.032$, -0.046 to 0.106) did not exclude zero. These are Fitzpatrick-associated observed-signal differences, not measured-melanin effects; device gain, contact pressure, anatomy, and protocol may confound them.

Under the historical session-anchored C0-versus-C2 comparison, per-band SBP MAE decreases in Fitzpatrick I–II ($7.865 \to 7.169$), III–IV ($8.151 \to 7.276$), and nominally V–VI ($9.042 \to 6.280$). The V–VI estimate is based on only six participants and is reported as indicative only; it cannot support a fairness or parity claim.

E. Independent Quality-Fusion and Guarded-Correction Test

Table VII reports pulse-rate results on 108 participants. Quality-weighted fusion achieved 3.690 BPM MAE versus 4.366 for visible and 5.037 for infrared alone. A post-hoc audit of the same stored channel estimates added an equal-weight comparator, with 3.985 BPM MAE. Quality-minus-comparator paired differences were -0.676 BPM (95% CI -1.210 to -0.159) versus visible, -1.347 (-1.872 to -0.880) versus infrared, and -0.294 (-0.567 to -0.069) versus equal weighting. These marginal intervals support an overall pulse-rate benefit in this dataset; they do not establish BP utility. For MST 7–10, the quality-minus-equal-weight difference was -0.402 BPM (-0.929 to 0.010), which does not establish subgroup benefit. The exploratory MST guard activated in 5% of outer folds and achieved 3.703 BPM; its corrected-minus-tone-blind difference was 0.013 (-0.010 to 0.053), with no established overall benefit.

F. VitalDB Context-Plus-PPG Frozen Holdout

The expanded analysis used 55 development subjects to freeze the estimators and fixed blend, followed by one evaluation on 228 cases from 227 disjoint held-out subjects and 5,472 post-calibration windows. Table VIII separates the large gain over retaining the early calibration value from the stricter incremental PPG contrast. Calibration plus time/demographic context accounted for most of the improvement. The frozen fixed 50/50 blend reduced SBP MAE from 15.827 to 15.375 mmHg ($\Delta = -0.451$, 95% CI -0.781 to -0.121) and DBP MAE from 9.189 to 8.833 mmHg ($\Delta = -0.357$, -0.512 to -0.194). It improved over context for 67.0% of subjects for SBP and 70.5% for DBP. For windows with ≥ 10 mmHg reference change, the incremental gains were 0.707 mmHg SBP (0.337–1.074) and 0.501 mmHg DBP (0.274–0.726). These results support a small incremental contribution of the PPG-augmented implementation within the VitalDB holdout, but absolute errors remain high and the test is retrospective and intraoperative rather than a prospective wearable validation.

G. State-of-the-Art Comparison

Table IX situates PC-AVCT against prior cuffless-BP results, organized around the axis we argue matters most for interpreting any number in this literature: what evaluation protocol produced it. Direct MAE comparison across datasets, populations, and BP ranges is only ever indicative – a point independently demonstrated by [1], which shows record-level leakage alone can inflate reported performance by more than 40 percentage points on the same data. We therefore report, for every comparator, whether the split was subject-independent, whether the setting was calibrated or calibration-free, and whether the study stratified by skin tone at all. Read narrowly, PC-AVCT's wearable-state repeated-validation session-anchored MAE (7.697/6.139 mmHg) is higher than the best calibrated

TABLE IX
State-of-the-Art Comparison: Cuffless/Calibrated PPG Blood-Pressure Estimation

Method	Dataset (N)	Split	Calib.	SBP	DBP	Skin-tone report
PPG2BP-Net [43]	Intraop. (4,185 subj.)	Subject-indep.	Required	0.209±7.509*	0.150±4.549*	Not reported
Calibration-free CV-dynamics [8]	UCI/MIMIC-II (225 subj.)	Subject-indep.	Free	7.41	3.32	Not reported
Benchmark (representative baseline) [1]	Four public PPG–BP datasets	Subject-level CV	Mixed	15.60 (Sensors)	–	Not reported; quantifies leakage inflation up to >40 pp
AVCT (companion work) [5]	Smartphone PPG	Single-point calib.	1 reading	2.05	1.67	Not reported
rPPG equity field study [24]	Nigeria, Fitzpatrick V/VI (306 enrolled; 249 analyzable)	Independent cuff ref.	Free (rPPG)	15.4	10.9	Yes – Fitzpatrick V/VI
PC-AVCT wearable-state (this work), session-anchored	AuroraBP (653 subj.)	3×5-fold participant CV, bootstrap CI	3 readings	7.697 [7.407, 8.008]	6.139 [5.899, 6.391]	Real Fitzpatrick I–VI; V–VI underpowered
PC-AVCT protocol-enriched upper bound	AuroraBP (653 subj.)	Same internal CV	3 readings	7.523 [7.243, 7.815]	5.743 [5.527, 5.977]	Protocol metadata limit transportability
PC-AVCT context-plus-PPG holdout	VitalDB (227 held-out subj.)	Locked subject holdout	3 windows	15.375	8.833	Not available; temporal test only

* Reported as mean error ± SD, not MAE; not directly comparable to the MAE figures in other rows.

results reported elsewhere and DBP is higher than the calibration-free result of [8]. Direct ranking is not justified because datasets, populations, calibration doses, reference protocols, and evaluation procedures differ. AuroraBP includes substantial within-session physiological changes, and our analysis excludes calibration-labelled rows and separates participants throughout outer validation. The VitalDB test supplies an independent-dataset evaluation of a separately developed implementation through a disjoint 227-subject holdout, but its intraoperative setting and high absolute errors still preclude clinical ranking or wearable-generalization claims. The appropriate conclusion is that the evidence is more stringently characterized, not that the method is superior to every comparator.

VI. Discussion

The primary AuroraBP comparison supports adding wearable-state context to the shared calibrated model, although posture and activity remain protocol annotations rather than independently estimated wearable states. The protocol-enriched arm achieves lower error but has greater dependence on acquisition metadata. Calibration dose materially affects that arm's performance. Because the common core already includes baseline BP, elapsed-time candidates, and a UCI prior, these results do not isolate PPG utility beyond a calibration-plus-context-only model on AuroraBP.

The independent studies answer different questions. The post-hoc equal-weight comparison supports quality-weighted pulse-rate fusion on the multi-wavelength dataset. VitalDB supports a small incremental BP contribution from a separately fitted context-plus-PPG implementation, combined with context through a fixed 50/50 blend. It does not test quality-dependent blending or transfer of AuroraBP-fitted parameters. The combined evidence supports calibrated tracking research but does not establish one universally generalizing architecture, an attractor-specific advantage, or clinical interchangeability. Pigmentation interventions remain secondary; small samples, possible indirect feature dependence, preclude fairness conclusions.

Clinical accuracy boundary. The method remains a research prototype for calibrated trend tracking. On the historical measurement-level standards analysis, SBP achieved BHS Grade C and failed the reported AAMI/ISO Criterion 1 check because error SD was 9.654 mmHg, above 8 mmHg. VitalDB absolute MAE was also high (15.375/8.833 mmHg). These failures cannot be repaired by emphasizing relative improvements, rejecting difficult observations after seeing their errors, or treating a calibrated tracking model as a validated cuff-replacement device. Existing data can support tail-error diagnostics and prospectively defined repeat-measurement rules, but not a claim of clinical interchangeability.

VII. Limitations

Small Fitzpatrick V–VI samples. The historical fixed holdout contains only 6 participants with Fitzpatrick V–VI labels; its fixed-model contrasts, Table S8, and the historical per-band C0/C2 comparison are exploratory. The strict repeated-CV cohort includes 25 V–VI participants, which is larger but still small relative to I–II ($n = 516$) and insufficient for a definitive equity claim. Oversampling, duplicating, or synthetically weighting these participants cannot create independent evidence. Consequently, pigmentation modeling is a secondary intervention audit rather than a headline validation claim, and the equity question cannot be resolved using the current datasets.

Synthetic pigmentation labels off AuroraBP. BIDMC, MIMIC, Welltory, and UCI contribute only a cross-dataset attractor-feature representation and prior; their pigmentation indices are permutation-derived synthetic ranks, not physiological measurements, and no equity claim in this paper rests on them. Single-site, in-lab protocol. AuroraBP measurements occur within < 1 h sessions in a controlled lab setting; the prior-anchored result (Section V-C) is therefore a same-session tracking result, not evidence about ambulatory, multi-day tracking, and we explicitly do not claim it as such. External evidence remains retrospective. The repeated AuroraBP analysis estimates internal stability because AuroraBP informed earlier feature and model decisions. The Appel–Theart analysis is independent but validates pulse extraction rather than BP tracking. VitalDB supplies a frozen, different-site, different-device test on 227 held-out subjects, but it remains a retrospective intraoperative cohort with invasive BP, no skin-tone labels, and a different use setting from ambulatory wearables. It is not a prospective clinical validation. Adaptive pipeline development. Repeated cross-validation protects each fitted model from direct participant overlap, but it does not erase optimism from choosing feature families, context arms, and post-processing strategies after working extensively with AuroraBP. The endpoint-specific analyses are therefore labelled exploratory, and their bootstrap intervals are not substitutes for evaluation on a new cohort. Multiplicity and analytical breadth. Holm adjustment is applied to the two retrospectively designated primary paired tests. The remaining ablations and subgroup comparisons are exploratory, with marginal intervals. This correction addresses the stated two-endpoint family; it does not account for the full history of model-development choices or substitute for prospective registration. Calibration burden. The primary model averages three initial cuff readings. In the protocol-enriched dose experiment, exactly one reading produces higher MAE (8.257/6.280 mmHg); the corresponding wearable-state dose comparison is not reported. The three-reading result should be interpreted as one-session repeated calibration, not one-point calibration. Context transportability. The protocol-enriched model uses recording duration, phase, and pressure-quality metadata that may proxy AuroraBP protocol structure. The conservative wearable-state arm removes those variables and remains beneficial, but assumes that posture and activity are available from a deployable wearable classifier. Warning-router sensitivity. The DBP warning router identifies a small subset with high precision but recalls only 4.3% of extreme-change rows at its selected threshold. It cannot be used to claim comprehensive failure detection, clinical safety, or permission to omit confirmatory cuff measurement. Measurement-level vs. participant-level reporting. Because each participant contributes $\sim$13 correlated measurements, uncertainty calculations that treat measurements as independent can be too narrow; measurement-weighted summaries also target a different estimand from participant-balanced summaries; we report both where possible and lead with participant-level bootstrap CIs for all primary claims. Pigmentation-correction operator is one design, not a proof of impossibility. Section III-B does not establish an overall benefit of the tested heuristic; its nonsignificant contrasts demonstrate neither equivalence nor impossibility of benefit. Quality fusion is not BP-fairness validation. The independent multi-wavelength result concerns pulse-rate extraction, and the available cuff value is cross-sectional. VitalDB has no pigmentation label. Neither result establishes that quality-aware fusion reduces BP error in Fitzpatrick V–VI or MST 7–10 participants. Derived clinical categories, not a classifier. Any ACC/AHA-stage confusion analysis in the released figures is a post-hoc discretization of a regression model's continuous output, not a trained classifier, and boundary-crossing small errors are penalized identically to large within-category errors. BHS/AAMI standards scope. Both grading systems were designed for cuff devices measuring absolute BP against an independent reference; applying them to a per-participant-calibrated change model is informative context, not compliance. Reproduction scope and feature attribution. All primary context and calibration-dose models were refitted from hashed feature-bank inputs in a recorded software environment, with outer-fold predictions, participant contrasts, feature selections, and gates saved. The historical fixed-estimator subgroup model was also refitted, and its new estimates replace the inherited subgroup values. The analysis rerun begins at feature-bank inputs; it does not repeat raw-waveform extraction. The UCI prior uses corrected features, so "no added Fitzpatrick terms" does not imply complete pigmentation independence. A calibration-plus-context-only AuroraBP comparator and an attractor-versus-standard-feature ablation remain necessary to isolate optical and attractor contributions.

VIII. Conclusion

PC-AVCT provides retrospective evidence for context-augmented session-calibrated BP tracking. The AuroraBP primary comparison supports added wearable-state context; the calibration-dose result belongs to the protocol-enriched arm. A separate post-hoc pulse-rate comparison supports quality-weighted fusion over equal weighting. Within VitalDB, a separately fitted context-plus-PPG implementation and fixed 50/50 blend provide small incremental BP improvements beyond calibration, time, and demographics, with high absolute errors. These results do not isolate quality-dependent BP blending, prove attractor-specific benefit, demonstrate transfer of the fitted AuroraBP model, validate AVCT/CST/ADT as physiological laws, or establish equitable, prospective, ambulatory, standards-compliant, or cuff-replacement performance.

Code and Reproduction Availability

The companion reproducibility package contains executable rerun and statistical-analysis scripts, preserved V26 helper functions, pinned dependencies, input SHA-256 hashes, model-selection manifests, and regenerated aggregate outputs. Locally retained row predictions and paired participant contrasts support result checking under the dataset access conditions. The public code archive excludes participant-level data. A public repository DOI remains to be assigned to the archived code release; the dataset DOI below identifies data, not this implementation.

Data Availability

AuroraBP is available through the restricted-access Zenodo record cited in [30] (DOI: 10.5281/zenodo.19099166). BIDMC and MIMIC-II are available through PhysioNet, and the UCI Cuff-Less Blood Pressure Estimation dataset is available through the UCI Machine Learning Repository. The multi-wavelength participant data are available through Mendeley Data [39]; VitalDB is openly available through PhysioNet [41]. Access conditions and citations are listed in Section IV-A. No directly identifying participant data are included in this manuscript or its figures.

Ethics Statement

This study is a secondary computational analysis of previously collected, deidentified datasets. The authors recruited no participants, performed no intervention, had no participant contact, and did not receive direct identifiers. Ethics approvals and consent procedures for the original data collections are reported by the respective dataset providers and source publications [26], [30], [32], [39]–[41]. No new participant consent was obtained for this secondary analysis. The manuscript materials available for this revision do not contain an institutional determination for this secondary analysis. Approval, exemption, or a determination that the analysis is not human-subjects research is therefore not asserted. This administrative status remains unresolved and must be documented by the authors before submission.

Author Contributions

Farouk Ganiyu Adewumi: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, and writing of the original draft. Timothy Oladunni: conceptualization, methodology, supervision, and review and editing of the manuscript. The stated roles are retained from the prior manuscript; both authors' approval of this revised version must be confirmed before submission.

Competing Interests

The authors declare no competing financial or nonfinancial interests.

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References

[1] S. González, W.-T. Hsieh, and T. P.-C. Chen, "A benchmark for machine-learning based non-invasive blood pressure estimation using photoplethysmogram," Scientific Data, vol. 10, p. 149, 2023.

[2] J. Kostrna, E. Oparina, C. Palacios, A. J. Rodriguez, J. Pei, A. Ajmal, and J. C. Ramella-Roman, "PPG-based heart rate accuracy in diverse populations: Investigating inequities across body composition and skin tones," arXiv preprint arXiv:2601.22377, 2026.

[3] S. Singh, M. R. Bennett, C. Chen, S. Shin, H. Ghanbari, and B. W. Nelson, "Impact of skin pigmentation on pulse oximetry blood oxygenation and wearable pulse rate accuracy: Systematic review and meta-analysis," Journal of Medical Internet Research, vol. 26, p. e62769, 2024.

[4] L. Osorio-Sanchez, J. M. May, and P. A. Kyriacou, "Evaluation of skin pigmentation effect on photoplethysmography signals using a vascular finger phantom with tunable optical and mechanical properties," Journal of Biomedical Optics, vol. 30, no. 11, p. 117002, 2025.

[5] T. Oladunni and F. G. Adewumi, "Attractor-vascular coupling theory: Formal grounding and empirical validation for AAMI-standard cuffless blood pressure estimation from smartphone photoplethysmography," arXiv preprint arXiv:2605.10871, 2026.

[6] M. Elgendi, R. Fletcher, Y. Liang, N. Howard, N. H. Lovell, D. Abbott, K. Lim, and R. Ward, "The use of photoplethysmography for assessing hypertension," npj Digital Medicine, vol. 2, p. 60, 2019.

[7] R. Mukkamala, J.-O. Hahn, O. T. Inan, L. K. Mestha, C.-S. Kim, H. Töreyin, and S. Kyal, "Toward ubiquitous blood pressure monitoring via pulse transit time: Theory and practice," IEEE Transactions on Biomedical Engineering, vol. 62, no. 8, pp. 1879–1901, 2015.

[8] H. Samimi and H. R. Dajani, "A PPG-based calibration-free cuffless blood pressure estimation method using cardiovascular dynamics," Sensors, vol. 23, no. 8, p. 4145, 2023.

[9] C.-F. Wang, T.-Y. Wang, P.-H. Kuo, H.-L. Wang, S.-Z. Li, C.-M. Lin, S.-C. Chan, T.-Y. Liu, Y.-C. Lo, S.-H. Lin, and Y.-Y. Chen, "Upper-arm photoplethysmographic sensor with one-time calibration for long-term blood pressure monitoring," Biosensors, vol. 13, no. 3, p. 321, 2023.

[10] M. Elgendi, F. Haugg, R. R. Fletcher, J. Allen, H. Shin, A. Alian, and C. Menon, "Recommendations for evaluating photoplethysmography-based algorithms for blood pressure assessment," Communications Medicine, vol. 4, p. 140, 2024.

[11] F. Takens, "Detecting strange attractors in turbulence," in Dynamical Systems and Turbulence, Warwick 1980, ser. Lecture Notes in Mathematics. Springer, 1981, vol. 898, pp. 366–381.

[12] J.-P. Eckmann, S. O. Kamphorst, and D. Ruelle, "Recurrence plots of dynamical systems," Europhysics Letters, vol. 4, no. 9, pp. 973–977, 1987.

[13] N. Marwan, M. C. Romano, M. Thiel, and J. Kurths, "Recurrence plots for the analysis of complex systems," Physics Reports, vol. 438, no. 5–6, pp. 237–329, 2007.

[14] J. S. Richman and J. R. Moorman, "Physiological time-series analysis using approximate entropy and sample entropy," American Journal of Physiology–Heart and Circulatory Physiology, vol. 278, no. 6, pp. H2039–H2049, 2000.

[15] C. Bandt and B. Pompe, "Permutation entropy: A natural complexity measure for time series," Physical Review Letters, vol. 88, no. 17, p. 174102, 2002.

[16] J. Allen, "Photoplethysmography and its application in clinical physiological measurement," Physiological Measurement, vol. 28, no. 3, pp. R1–R39, 2007.

[17] C. Hörandtner, M. Bachler, W. Sehnert, I. Mikisek, T. Mengden, S. Wassertheurer, and C. C. Mayer, "Attractor reconstruction for quantifying the arterial pulse wave morphology during device-guided slow breathing," Cardiovascular Engineering and Technology, vol. 13, no. 6, pp. 939–949, 2022.

[18] T. Oladunni and F. G. Adewumi, "Cardiac stability theory: An axiomatically grounded framework for continuous cardiac health monitoring via smartphone photoplethysmography," arXiv preprint arXiv:2604.23876, 2026.

[19] T. Oladunni and F. G. Adewumi, "Attractor domain theory: A mathematical framework for cardiovascular attractor analysis with wearable photoplethysmography (PPG) validation," arXiv preprint arXiv:2606.22039, 2026.

[20] C. E. Shannon, "A mathematical theory of communication," Bell System Technical Journal, vol. 27, no. 3, pp. 379–423, 1948.

[21] T. M. Cover and J. A. Thomas, Elements of Information Theory, 2nd ed. Wiley-Interscience, 2006.

[22] R. Al-Halawani, M. Qassem, and P. A. Kyriacou, "Analysis of the effect of skin pigmentation and oxygen saturation on monte carlo-simulated reflectance photoplethysmography signals," Sensors, vol. 25, no. 2, p. 372, 2025.

[23] R. Jakachira, W. Yan, S. C. Thomas, Y. Macias-Sanchez, L. Werner, J. A. Burrow, S. Dunsiger, and K. C. Toussaint, Jr., "Evaluation of a polarization-sensitive, dual-wavelength wearable photoplethysmography sensor across a range of skin tones," Biophotonics Discovery, vol. 3, no. 1, p. 012509, 2026.

[24] D. Dasa and P. Davies, "Skin tone and diagnostic equity in contactless blood pressure screening: A prospective observational field evaluation of remote photoplethysmography in nigeria," BMJ Open, vol. 16, no. 6, p. e119311, 2026.

[25] C. MacIsaac, M. Nguyen, A. Uy, T. Kong, and A. Hedayatipour, "A programmable gain calibration method to mitigate skin tone bias in PPG sensors," Biosensors, vol. 15, no. 7, p. 423, 2025.

[26] R. Mieloszyk, H. Twede, J. Lester, J. Wander, S. Basu, G. Cohn, G. Smith, D. Morris, S. Gupta, D. Tan, N. Villar, M. Wolf, S. Malladi, M. Mickelson, L. Ryan, L. Kim, J. Kepple, S. Kirchner, E. Wampler, R. Terada, J. Robinson, R. Paulsen, and T. S. Saponas, "A comparison of wearable tonometry, photoplethysmography, and electrocardiography for cuffless measurement of blood pressure in an ambulatory setting," IEEE Journal of Biomedical and Health Informatics, vol. 26, no. 7, pp. 2864–2875, 2022.

[27] I. Sanches, V. V. Gomes, C. Caetano, L. S. B. Cabrera, V. H. Cene, T. Beltrame, W. Lee, S. Baek, and O. A. B. Penatti, "MIMIC-BP: A curated dataset for blood pressure estimation," Scientific Data, vol. 11, p. 1233, 2024.

[28] F. M. Dias, D. A. C. Cardenas, M. A. F. Toledo, F. A. C. Oliveira, E. Ribeiro, J. E. Krieger, and M. A. Gutierrez, "Exploring the limitations of blood pressure estimation using the photoplethysmography signal," Physiological Measurement, vol. 46, no. 4, p. 045007, 2025.

[29] S. Mehta, N. Kwatra, M. Jain, and D. McDuff, "Examining the challenges of blood pressure estimation via photoplethysmogram," Scientific Reports, vol. 14, p. 18318, 2024.

[30] Microsoft Research, "Aurora-BP Study Dataset," Zenodo, 2022, restricted-access research dataset.

[31] M. A. F. Pimentel, A. E. W. Johnson, P. H. Charlton, and D. A. Clifton, "BIDMC PPG and Respiration Dataset," PhysioNet, 2018, version 1.0.0.

[32] M. A. F. Pimentel, A. E. W. Johnson, P. H. Charlton, D. Birrenkott, P. J. Watkinson, L. Tarassenko, and D. A. Clifton, "Toward a robust estimation of respiratory rate from pulse oximeters," IEEE Transactions on Biomedical Engineering, vol. 64, no. 8, pp. 1914–1923, 2017.

[33] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," Circulation, vol. 101, no. 23, pp. e215–e220, 2000.

[34] M. Saeed, M. Villarroel, A. T. Reisner, G. Clifford, L.-W. Lehman, G. Moody, T. Heldt, T. H. Kyaw, B. Moody, and R. G. Mark, "Multiparameter intelligent monitoring in intensive care ii (MIMIC-II): A public-access intensive care unit database," Critical Care Medicine, vol. 39, no. 5, pp. 952–960, 2011.

[35] M. Saeed, M. Villarroel, A. T. Reisner, G. Clifford, L.-W. Lehman, G. Moody, T. Heldt, T. Kyaw, B. Moody, and R. Mark, "MIMIC-II Clinical Database," PhysioNet, 2011, version 2.6.0.

[36] A. Neshitov, K. Tyapochkin, E. Smorodnikova, and P. Pravdin, "Wavelet analysis and self-similarity of photoplethysmography signals for HRV estimation and quality assessment," Sensors, vol. 21, no. 20, p. 6798, 2021.

[37] M. Kachuee, M. M. Kiani, H. Mohammadzade, and M. Shabany, "Cuff-Less Blood Pressure Estimation," UCI Machine Learning Repository, 2015.

[38] M. Kachuee, M. M. Kiani, H. Mohammadzade, and M. Shabany, "Cuff-less high-accuracy calibration-free blood pressure estimation using pulse transit time," in 2015 IEEE International Symposium on Circuits and Systems (ISCAS), 2015, pp. 1006–1009.

[39] A. Appel and R. Theart, "Photoplethysmography device data: Hardware and participant study insights," Mendeley Data, 2025.

[40] H.-C. Lee, Y. Park, S. B. Yoon, S. M. Yang, D. Park, and C.-W. Jung, "VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients," Scientific Data, vol. 9, p. 279, 2022.

[41] H.-C. Lee and C.-W. Jung, "VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients," PhysioNet, 2022, version 1.0.0.

[42] S. Holm, "A simple sequentially rejective multiple test procedure," Scandinavian Journal of Statistics, vol. 6, no. 2, pp. 65–70, 1979.

[43] J. Joung, C.-W. Jung, H.-C. Lee, M.-J. Chae, H.-S. Kim, J. Park, W.-Y. Shin, C. Kim, M. Lee, and C. Choi, "Continuous cuffless blood pressure monitoring using photoplethysmography-based PPG2BP-net for high intrasubject blood pressure variations," Scientific Reports, vol. 13, p. 8605, 2023.

Appendix A
Supplementary Implementation Details

A. Multi-Wavelength Pulse Processing

For each device, visible and infrared pulse rates are estimated around the scheduled reference time using a window of up to 12 s, with at least 7.5 s and 150 samples required. Timestamps are converted to seconds, sorted, and deduplicated; signals are interpolated to a uniform grid using the median interval, with inferred sampling rate clipped to 20–500 Hz. A third-order 0.55–4 Hz Butterworth filter is applied forward and backward after detrending. Welch spectra use up to 8 s segments (at least 256 samples when available), with 50% overlap. The dominant frequency is sought in 0.65–3.5 Hz; spectral concentration c is the fraction of pulse-band power within 0.18 Hz of that peak.

An autocorrelation pulse estimate searches lags corresponding to 39–210 BPM. Let a be the selected autocorrelation peak clipped to $[0, 1]$, and h_F, h_A the spectral and autocorrelation rates. When $|h_F - h_A| \leq 15$ BPM, the channel estimate is $(ch_F + ah_A)/(c + a)$; otherwise the estimate with the larger reliability component is used. Channel quality is

$$q = \mathrm{clip}\left(\sqrt{ca}\, g, 10^{-4}, 1\right), \tag{7}$$

$$g = \begin{cases} 1, & |h_F - h_A| \leq 15, \\ e^{-|h_F - h_A|/30}, & \text{otherwise.} \end{cases} \tag{8}$$

The tone-blind fusion estimate is $(q_V h_V + q_I h_I)/(q_V + q_I)$. The optional tone-dependent arm replaces the weights by $q_V e^{-\alpha z}$ and $q_I e^{\alpha z}$, with $z = \mathrm{clip}[(\mathrm{MST} - 1)/9, 0, 1]$. Candidate α values are $\{0, \pm0.25, \pm0.5, \pm0.75, \pm1, \pm1.5, \pm2, \pm3\}$. Four participant folds within each outer training set score these fixed candidate rules. The selected nonzero exponent must improve mean fold MAE by at least 0.10 BPM and improve every inner fold; otherwise $\alpha = 0$. The seed is 20260827, with repeat/fold offsets defined in the companion script. These methods estimate pulse rate, not BP.

B. VitalDB Extraction and Model Selection

Eligible cases have all three required tracks and a case duration of at least 1,800 s. Cases are deterministically shuffled with seed 20260827; positions 1–60 define development and 61–300 define the candidate holdout. The saved audit contains 55 development cases/subjects and 228 eligible holdout cases from 227 subjects, with no subject overlap. Eligibility and sampling are retrospective, and selecting windows across the completed record is not an online scheduling policy.

Candidate centers start no earlier than 120 s, proceed in 30-s increments, and require at least 900 s of usable overlapping record. PPG is retained at approximately 100 Hz. A 20-s window must contain at least 75% of the expected samples and 85% finite values; gaps are interpolated. Values are clipped to the median plus/minus eight robust SD estimates, detrended, and filtered with a third-order 0.5–8 Hz bandpass (upper cutoff limited by sampling rate). Pulse quality is spectral concentration multiplied by $\exp[-\min(\text{interval CV}, 2)]$; windows below 0.04 are excluded. Pulse frequency search, spectral concentration, and beat-interval limits follow the companion extraction code. This quality filter is held fixed, but a no-quality-filter ablation is not reported.

Reference BP is the median of at least three plausible values within ±5 s of the center. SBP must lie in 70–220 mmHg and DBP in 35–130 mmHg; local robust SD must not exceed 8 and 6 mmHg, respectively. Pulse pressure must lie in 20–130 mmHg. These reference-based exclusions do not use prediction errors but restrict the evaluated BP distribution. Cases require at least eight valid windows. The first three define median BP and feature anchors; up to 24 later windows are selected using evenly spaced indices over the valid future windows.

The context model uses anchor BP, $\log(1 + \text{elapsed seconds})$, age, sex, and preoperative hypertension (missing hypertension is coded as zero). The PPG model adds current and calibration-relative optical features listed by the columns function in the companion script. Candidate estimators comprise standardized ridge regression with 13 logarithmically spaced penalties from 10^{-3} to 10^{3} and three absolute-loss gradient-boosting configurations. Development-only five-fold participant validation selects estimators, followed by a fixed blend weight from $\{0, 0.25, 0.5, 0.75, 1\}$; a nonzero weight requires at least 0.10 mmHg improvement over context. Development selection is not independently nested across the estimator and blend choices, but holdout observations are excluded from both. Delta predictions are clipped at training-target 1st/99th percentiles.

Both endpoints selected ridge context models with penalty approximately 31.6228. The SBP PPG model uses absolute-loss boosting, learning rate 0.03, 15 leaves, regularization 5, 250 iterations, and minimum leaf size 20; the DBP PPG model uses ridge with penalty approximately 31.6228. Both selected a constant 0.5 PPG weight. The saved predictions satisfy this fixed-blend identity to numerical precision. Errors are averaged over all windows from all cases of a subject before subject bootstrap; the subject contributing two cases remains one bootstrap unit.

C. Audit Scope and Reproduction Status

The independent-dataset saved-result audit uses 10,000 participant bootstrap resamples with seed 20260910 for the pulse comparisons and verifies window uniqueness, case/subject counts, development/holdout separation, the fixed VitalDB blend, and the pulse-fusion equation. The AuroraBP analysis was separately rerun: the context arms and all three calibration doses were refitted across three repeats of five participant folds. The package records hashed feature inputs, software versions, selected features and gates, row-level outer-fold predictions, and paired participant contrasts. The historical subgroup model was also refitted. The rerun starts from feature banks; raw-waveform extraction is outside its scope.

Appendix B
Historical Development and Auxiliary Analyses

The following analyses preserve the development record. They are secondary or exploratory. The final subgroup subsection reports newly refitted estimates; development codes C0–C4 refer to the historical arms.

A. Exploratory Endpoint-Specific Calibration and Warning Router

Table S2 reports the post-primary endpoint analysis. A nested, tightly clipped residual calibrator reduced SBP participant-level MAE from 7.611 to 7.555 mmHg, a paired improvement of 0.056 mmHg (95% CI 0.020–0.091; 55.9% of participants helped). The absolute gain is small, and tail behavior changed little (P90 absolute error 16.862 → 16.820 mmHg; P95 22.184 → 22.120 mmHg). DBP candidate specialization did not improve inner-fold performance consistently, so the base prediction was retained at 5.784 mmHg. This endpoint-specific policy is more conservative than adopting one post-processing rule for both targets. The exploratory DBP router was therefore frozen as warning-only. Extreme DBP change prevalence was 20.5% (95% CI 19.1–21.9%). The router achieved ROC AUC 0.794 (0.778–0.809), average precision 0.517 (0.483–0.555), and Brier score 0.130 (0.122–0.137). At the nested operating threshold, warning precision was 0.773 (0.695–0.854), but recall was only 0.043 (0.035–0.052) and the warning rate was 1.1% (0.9–1.4%). The router is thus a rare high-precision flag at the evaluated threshold; it is not a screen for most extreme changes, does not reduce reported DBP MAE, and must not be interpreted as a clinical safety mechanism.

B. Historical Duplicate-Reading Noise Diagnostic

Table S3 retains a historical duplicate-reading diagnostic, not a universal achievable-error bound. Estimating per-reading noise as $\sigma_{P-S}/\sqrt{2}$ assumes independent, equal-variance observer errors. The supplied code distinguishes a stored average of two readings from a single-reading target and propagates target and anchor errors for a legacy two-time-point difference. Its 1.754/2.412 mmHg Gaussian absolute-noise estimates and 2.6%/9.2% variance ratios therefore must not be treated as validated floors for the primary three-reading anchor. Shared observer error, physiological change during measurement, and the actual target-averaging rule can change the estimates. A primary-endpoint noise calculation would require those components and their covariances; subtracting a putative floor from model MAE does not measure recoverable physiological error.

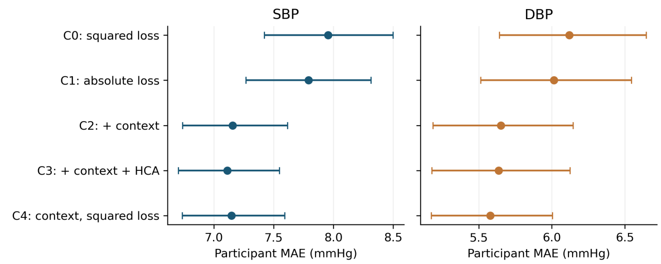

Fig. S1. Historical fixed-split session-anchored ladder with 95% participant-bootstrap CIs. This figure documents model development; Table II provides the primary repeated participant-validation result.

C. Recalibration Cadence

The historical code evaluated nominal recalibration intervals of 1, 3, 6, 12, and 24 h, with nearly identical MAE (8.593–8.594 mmHg SBP; 6.020–6.022 mmHg DBP). Those labels do not establish distinct effective recalibration schedules in the analyzed short laboratory sessions. The stored comparison lacks an event-count and elapsed-time audit sufficient to interpret a cadence effect. We therefore withdraw the inference that cadence beyond one hour is immaterial and make no recommendation about recalibration frequency from this analysis.

D. Multi-Window Feature Averaging

Averaging attractor features over multiple 10 s windows per recording (up to 3 available) was hypothesized to reduce measurement noise (Table S4 shows within/between SD ratios up to approximately 0.45). It did not help: holdout MAE with multi-window averaging (6.393/4.828 mmHg SBP/DBP) was statistically indistinguishable from, and nominally slightly worse than, a matched single-window configuration (6.346/4.803 mmHg), both close to the legacy single-window baseline (6.315/4.789 mmHg). We report this as a negative result (Table S1).

E. Historical Fixed-Split Session-Anchored Ladder

During earlier development, the model-development ladder was evaluated on a fixed 501/168 participant split. Table S5 preserves that analysis for transparency: C4 reduced MAE from 8.856/6.582 to 7.147/5.579 mmHg. Because the same 168 participants were examined across successive ablations, these values are historical development evidence and are not the paper's primary validation result. The corresponding repeated participant-validation estimates in Table II are modestly worse (7.523/5.743 mmHg for protocol-enriched context), consistent with mild optimism from the fixed split rather than a collapse of the effect.

F. Calibration-Row Leakage Audit

Table S6 reports all four checks from Section III-E. Check A shows calibration-flagged rows are smaller in mean $|\Delta|$ than unflagged rows (6.774 vs. 9.902 mmHg

TABLE S1
Interventions Tested and Not Adopted (Elimination Log)

Intervention	Times tested	Outcome
Absolute-error loss (vs. squared)	3	Never CI-supported: +0.152 [−0.001, 0.311] (51.8% helped); +0.102 [−0.028, 0.240]; +0.163 [−0.008, 0.345] (exactly 50.0% helped).
HCA feature augmentation	2	No CI-supported benefit: −0.077 [−0.183, 0.024]; +0.043 [−0.026, 0.115] (53.0% helped).
Signal-level pigmentation (Beer–Lambert) correction	2	First variant increased error; the repeated-CV corrected channel changed MAE by −0.007 [−0.042, 0.029] SBP and −0.030 [−0.062, 0.003] DBP – no established overall benefit (Section III-B).
Demographic (Fitzpatrick-band) rebalancing / group gating	5	Consistently harmful: e.g., -0.291 $[-0.420, -0.170]$ mmHg SBP; widens between-band gap in every run (Section V-D).
Multi-window feature averaging	1	No measurable benefit; nominally worse (Section B-D).
Nested gate/threshold search (early pipeline)	1	Higher error than a plainer reference baseline (8.253/8.183 vs. 8.109 mmHg SBP).
Passed sanity check: metadata-only permutation control	1	AUC = 0.492 (≈ chance) – is consistent with chance under this permutation control; does not exclude other confounds.

TABLE S2
Exploratory Endpoint-Specific Nested Analysis (653 Participants)

Target	Arm	MAE	P90 AE	P95 AE
SBP	Base	7.611	16.862	22.184
SBP	Clipped residual	7.555	16.820	22.120
DBP	Base	5.784	12.043	15.630
DBP	Selected policy	5.784	12.043	15.630

SBP paired gain: 0.056 mmHg [0.020, 0.091]; DBP gain: 0.000.

TABLE S3
Historical Duplicate-Reading Diagnostic (8,918/8,916 SBP/DBP pairs)

Quantity (mmHg)	SBP	DBP
σ(primary − secondary reading)	3.108	4.275
σ_{cuff} (per-reading noise)	2.198	3.023
Legacy Gaussian absolute-noise estimate	1.754	2.412
Observed SD of Δ	13.70	9.98
Legacy noise/observed variance ratio	2.6%	9.2%
Residual SD under legacy assumptions	13.52	9.51
Early reference model MAE	8.11	–

TABLE S4
Historical Within- and Between-Recording Feature SD

Feature	Within SD	Between SD	SD ratio
σ_m	0.0046	0.0102	0.449
CSI	0.0043	0.0138	0.311
Recurrence rate	0.0040	0.0130	0.305
Kurtosis	0.0902	0.3086	0.292
RQA entropy	0.0734	0.2520	0.291
Skewness	0.0386	0.1389	0.278
Sample entropy	0.0133	0.0527	0.252
Determinism	0.0114	0.0475	0.240
Spectral entropy	0.0092	0.0488	0.189
Permutation entropy	0.0028	0.0190	0.150

TABLE S5
Historical Fixed-Split Session-Anchored Ladder (168 Evaluation Participants)

Arm	SBP MAE [95% CI]	DBP MAE [95% CI]
Calibration retention	8.856	6.582
C0 legacy (squared)	7.956 [7.424, 8.501]	6.120 [5.641, 6.647]
C1 absolute loss	7.793 [7.270, 8.316]	6.014 [5.513, 6.546]
C2 +context	7.156 [6.740, 7.618]	5.651 [5.186, 6.146]
C3 +context+HCA	7.113 [6.703, 7.550]	5.636 [5.178, 6.126]
C4 +context (squared)	7.147 [6.736, 7.593]	5.579 [5.174, 6.004]
Reduction, calib. → C4	1.709 (19.3%)	1.003 (15.2%)

SBP, development set) but not trivially near-zero, and the fraction of rows within 1 mmHg is nearly identical between flagged and unflagged rows (9.8% vs. 8.6%). The decisive Check D – flag removed from features, flagged rows excluded from both fit and scoring – yields a context gain of +0.614 mmHg SBP $[0.346, 0.889]$ and +0.513 mmHg DBP $[0.268, 0.743]$, both CI-supported and close to the unaudited historical figures. Leave-one-out attribution (Figure S2) shows larger predictive contributions from recording duration and post-exercise activity, not the calibration flag (LOO damage for the flag: 0.0000 SBP / 0.0027 DBP). One honest qualification: the single largest contributor to the context gain is recording duration (0.244 mmHg SBP / 0.366 mmHg DBP of leave-one-out damage), which is an acquisition property (likely proxying protocol step or signal stability) rather than a purely physiological state variable. "Context = posture and activity" is only partly right; post-exercise activity (0.154/0.144 mmHg) is the clearest genuinely physiological contributor, with posture third.

G. Clinical Standards Context

Table S7 reports BHS grading and AAMI/ISO 81060-2 Criterion 1 for the C4 session-anchored model. The model improves cumulative within-5/10/15 mmHg percentages over both calibration retention and the legacy baseline for both SBP and DBP, and reaches BHS Grade B for DBP (Grade C for SBP); it passes AAMI Criterion 1 for DBP (mean error 0.645, SD 7.992 $\leq$ 8 mmHg) but not for SBP (SD 9.654 $>$ 8 mmHg). We report both gradings as indicative context, not a validation claim: BHS and AAMI/ISO were written for cuff devices measuring

TABLE S6
Calibration-Row Leakage Audit (168 holdout participants)

Check	SBP gain [CI]	DBP gain [CI]	Verdict
B (flagged excl.)	0.741 [0.440, 1.062]	0.606 [0.372, 0.854]	Supp.
C (flag dropped)	0.809 [0.501, 1.122]	0.538 [0.309, 0.770]	Supp.
D (strict)	0.614 [0.346, 0.889]	0.513 [0.268, 0.743]	Supp.

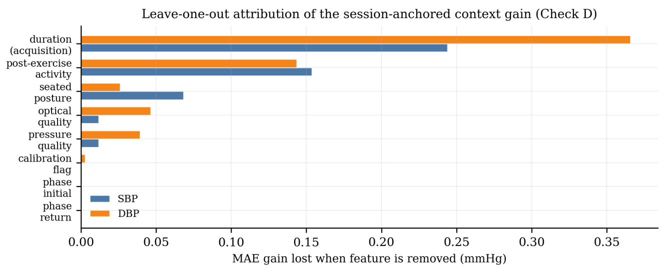

Fig. S2. Leave-one-out attribution of the session-anchored context gain (Check D). The largest contributor is recording duration – an acquisition property, not a purely physiological one – followed by post-exercise activity.

absolute BP against an independent reference, and this model is calibrated per participant.

H. What Did Not Work

Table S1 summarizes every intervention tested and rejected in this study, alongside one passed sanity check. We regard this table as carrying as much evidentiary weight as Table S5: a result that survives repeated, adversarially-minded attempts to break it is more credible than one presented without them.

I. Refitted Fixed-Estimator and Primary-Cohort Subgroups

The historical prior-anchored estimator was refitted under the recorded environment, and Table S8 reports its newly generated predictions. These estimates supersede the earlier rounded subgroup values; they do not assert bitwise recovery of the original run. The companion change record retains the old-value comparison. The same estimator is applied within each analysis before predictions are grouped by Fitzpatrick band. The primary wearable-state summary uses the independently regenerated repeated-CV predictions, not the historical fixed split. These are exploratory marginal intervals. Differences in sample size, anchoring, and validation design prevent direct comparison between the two analysis blocks. In particular, the small V–VI samples do not establish parity or equivalence, irrespective of the ordering of the point estimates.

TABLE S7
BHS Grading and AAMI/ISO 81060-2 Criterion 1
(Session-Anchored C4 Model, Measurement-Level)

Target/Arm	≤5mm	≤10mm	≤15mm	Grade	AAMI-1
SBP calibration	45.9%	72.3%	83.9%	D	Fail
SBP baseline (C0)	45.8%	73.8%	86.6%	C	Fail
SBP model (C4)	48.0%	75.1%	89.7%	C	Fail
DBP calibration	56.0%	82.9%	92.2%	B	Fail
DBP baseline (C0)	56.0%	83.1%	93.6%	B	Fail
DBP model (C4)	58.7%	85.6%	94.9%	B	Pass

TABLE S8
Regenerated Participant MAE by Fitzpatrick Band

Analysis	Target	Band (n)	MAE	95% CI
Prior refit	SBP	I–II (133)	5.837	[5.513, 6.178]
Prior refit	SBP	III–IV (29)	5.568	[4.703, 6.549]
Prior refit	SBP	V–VI (6)	5.341	[3.847, 6.841]
Prior refit	DBP	I–II (133)	4.439	[4.118, 4.795]
Prior refit	DBP	III–IV (29)	4.781	[3.674, 6.370]
Prior refit	DBP	V–VI (6)	4.020	[2.931, 5.197]
Primary CV	DBP	I–II (516)	6.078	[5.805, 6.362]
Primary CV	DBP	III–IV (112)	6.435	[5.862, 7.045]
Primary CV	DBP	V–VI (25)	6.053	[5.065, 7.122]
Primary CV	SBP	I–II (516)	7.680	[7.355, 8.022]
Primary CV	SBP	III–IV (112)	7.786	[7.050, 8.610]
Primary CV	SBP	V–VI (25)	7.646	[6.052, 9.471]